MIL-G-80239B
24 August 1987
SUPERSEDING
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MILITARY SPECIFICATION

GRINDING MACHINES, NON-TRAVERSING ROTARY TABLE, VERTICAL SPINDLE, UNIVERSAL, MANUAL, AUTOMATIC, AND COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, vertical spindle, rotary table, manual, automatic cycle, and computer numerically controlled (CNC) universal grinding machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following types, classes, styles, and sizes. The type, class, style, and size to be furnished shall be as specified (see 6.2.1).

Type I - Operator controlled (manual).
Type II - Automatic cycle controlled.
Type III - Computer numerically controlled (CNC).

Class 1 - Stationary cross rail.
Class 2 - Vertically adjustable cross rail.

Style A - Single grinding head.
Style B - Double grinding head.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Size	Worktable diameter* (inches)	Work swing (inches)	Maximum required hole depth ground (inches)
1	24	36	18
2	36	44	18
3	48	60	24
4	60	68	24

*Optional worktable diameters available (in 6" increments).

2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

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U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification as specified. Unless otherwise specified, DoD-adopted documents shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 6 - Enclosures for Industrial Controls and Systems.

ANSI/NFPA 70 - National Electrical Code.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B7.1 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels.

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines.

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels.

ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

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JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials, and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One Inch Perforated Tape.

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

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RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 - Construction Standards for Numerical Machine Tool Control.

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006-5009.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. If the text of this specification conflicts with the cited references (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design. The machine shall be new and one of the manufacturer's current models or a modification of a current model, as specified by the procuring activity (see 1.2). Type I machines shall be manually controlled, requiring the operator to perform all machine functions. Type II machines shall be automatic cycle controlled, performing all machine functions after the operator performs loading and machine set-up duties. Type III machines shall be controlled by solid-state, soft-wired, integrated circuit computer numerical control (CNC) systems. Class 1 machines shall have a stationary cross rail supporting the cross slide at a fixed height above the worktable; class 2 machines shall have a column(s) to allow for vertical adjustment of the cross rail above the worktable. Style A machines shall have a single grinding head and style B machines shall have two grinding heads, either characteristically identical or different, dependent upon the application, as specified by the procuring activity (see 3.3.7). The machine shall be capable of precision finishing. The machine shall grind inside and outside diameters, cylindrical surfaces, tapered surfaces, shoulders, and faces in one set-up of the workpiece in accordance with the requirements given in 3.5. Machine design shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.1.1 Measurement system. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.1.1.1 Measuring and indicating device calibrations. Unless otherwise specified (see 6.2.1), either US or SI units shall be used to graduate measuring and indicating devices. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.1.1.2 Digital readout system. Unless otherwise specified (see 6.2.1), a digital readout (DRO) system shall be provided to reference the wheelhead position (Z and X axes) on type I and type II machines. Type III machine CNC systems shall be provided with a data display (see 3.3.13.1). The DRO system shall consist of an electronic display for each axis and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The DRO resolution shall be 0.00005 inch. The system accuracy shall be 0.0002 inch and the positioning repeatability shall be 0.0001 inch. Special requirements for DRO system components shall be as specified (see 6.2.1). The readout system shall indicate axis location in both the US and SI systems of measurements.

3.1.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and inadvertent damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis (installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops). Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination of oil, dust, chips, coolants, and airborne particles.

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3.1.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual digits not less than 0.56 inch high. The console shall display at least six digits simultaneously. The display console shall also provide keyboard entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and installed so as not to interfere with the normal operation of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.1.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The materials shall have been relcaimed in a manner which will restore them to the originally selected for use on the machine.

3.1.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.1.4 Operator's controls. Each machine shall be provided with an oil-tight overhead pendent or push button station located convenient to the normal operator position. Unless otherwise specified (see 6.2.1), the control shall have at least the following features:

Grinding wheel start/stop (rotation)
Table start/stop/jog/reverse (rotation)
Grinding head up/down/jog
Coolant start/stop
Emergency stop (entire machine)
Slide (horizontal, vertical, angle) jog

Type II machines shall also have a "Cycle start/stop" feature (see 3.3.12). Type III machine controls shall also meet the CNC requirements given in 3.3.13.

3.1.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the grinding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and OSHA 29 CFR 1910 that

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are applicable to the machine itself. In the event of a conflict between the requirements of ANSI B11.9 and OSHA 29 CFR 1910, the requirements of OSHA 29 CFR 1910 shall apply.

3.1.6 Mercury restriction. The machine shall not contain mercury nor be exposed to free mercury during manufacture.

3.1.7 Asbestos restriction. Asbestos shall not be used on or in the machine.

3.1.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.1.9 Interchangeability. Replacement parts (spare or repair parts) shall be manufactured to definite dimensions and tolerances. Tooling and accessories normally supplied as standard equipment shall also be replaceable. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. All parts subject to future replacement as spare or repair parts shall be assigned a part number identified in the repair parts manual. Foreign sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from a domestic source.

3.2 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solders or pastes shall be used for reclaiming any defective part.

3.2.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.2.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.2.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.2.6 Threads. All threaded parts, either US or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.2.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.0002 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.8 Machine ways. Machine guideways shall be adequately protected regardless of machine component position to exclude hazards that may cause damage or result in inaccurate functioning of any members. Machine ways shall be hardened and ground and suitable for precision grinding machine applications required herein.

3.2.9 Slides. All sliding members shall be provided with gibs, guides, or other means to maintain the alignment and accuracy tolerances required herein for the machine type specified.

3.2.10 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.2.11 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.3 Components. Each machine shall consist of not less than a base, rotary worktable, cross rail, cross slide, grinding wheelhead assembly, wheel dresser, drive mechanisms, and controls. In addition, class 2 machines shall have a column(s) to support a vertically adjustable cross rail and a cross rail elevating mechanism. Style B machines shall have two grinding wheelhead assemblies. Any additional components necessary for the operation of the machine shall also be furnished in accordance with the manufacturer's standard practice.

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3.3.1 Base. The base shall be of epoxy granite, a casting of iron or iron alloy, or a fabricated steel weldment, ribbed and braced to minimize distortion and withstand forces imposed when the machine is utilized to its maximum capacities. The base shall support or house all major machine components and related parts. Means shall be provided to accept a stationary cross rail on class 1 machines and a column(s) on class 2 machines to accept a traveling cross rail for mounting the grinding wheelhead assembly. Means shall also be provided to support the work spindle unit for the rotating worktable. Guarding shall be located around the top of the base to enclose the work spindle and rotating worktable. Hydraulic and coolant reservoirs, when provided, shall be separate external units. Means shall be provided to collect and discharge coolant and grinding swarf from the base. An access door(s) or removable plate(s) shall be provided, as required, for servicing components housed within the base. The base shall have means for leveling and securing the machine to the floor.

3.3.2 Worktable. The worktable shall be a rotary table capable of supporting a workpiece while maintaining the rotational balance required to meet the performance and accuracy requirements herein. The top and peripheral surfaces of the rotating worktable shall be finished to meet the alignment tolerances given in table II. Unless otherwise specified (see 6.2.1), worktable T-slots, their number, and configuration shall be manufacturer's standard. Means shall be provided to securely clamp the workpiece on the rotating worktable and eliminate workpiece slippage up to the maximum spindle drive rotation. Type I machine worktable rotation shall be controlled by the operator; type II machine worktable rotation shall be controlled by the automatic cycle; and type III machine worktable rotation shall be controlled by the CNC system. Work spindle speeds, speed range, worktable diameter, and motor horsepower shall meet the requirements given in table I.

3.3.2.1 Worktable spindle. The worktable shall rotate on a self-contained work spindle supported by the base, held in alignment by ball or roller bearings. The work spindle shall be independent, driven by an infinitely variable speed electric motor or hydraulic drive through V-belts. The work spindle shall be steel alloy with its nose and worktable surfaces hardened and ground. The work spindle nose shall be manufacturer's standard design, arranged for convenient and accurate attachment of the rotating worktable and faceplates. Machine controls shall be provided for starting, stopping, and reversing work spindle rotation and adjusting spindle RPM throughout the available work spindle speed range.

3.3.3 Cross rail. Each machine shall have a cross rail to support the cross slide and grinding wheelhead assembly. The cross rail shall be of epoxy granite, a casting of iron or iron alloy, or a fabricated steel weldment. Class 1 machines shall have a stationary cross rail integral with or securely attached to the base to support the cross slide at a fixed position above the rotary worktable. Class 2 machines shall have a vertically adjustable

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cross rail mating with vertical column ways to support the cross slide at varying heights above the rotary worktable. The cross rail shall have guideways for the precision horizontal movement of the cross slide. The cross rail shall support or house the cross slide, grinding wheelhead assembly, cross slide feed mechanisms, and related components. On class 2 machines, the cross rail shall accept components of the cross rail elevating mechanism (see 3.3.3.1). Means shall be provided to automatically clamp the cross slide in place after positioning.

3.3.3.1 Cross rail elevating mechanism. Class 2 machines shall have a cross rail elevating mechanism to raise and lower the cross rail and related components to any position above the worktable within the limits of its travel. Unless otherwise specified (see 6.2.1), the elevating mechanism shall be enclosed in a housing, mounted on the top of both columns, straddling the base and forming a bridge over the base on two column machines. Manual adjustment and power traverse of the cross rail shall be provided on all machines. Cross rail vertical travel shall be operator controlled. When specified (see 6.2.1), type III machine cross rail elevation shall be controlled by the CNC system. A braking mechanism shall be provided on the cross rail elevating mechanism.

3.3.4 Columns. Class 2 machines shall have either one or two vertical columns. Each column shall be of epoxy granite, iron, or steel, ribbed and braced to support the traveling cross rail, cross slide, and grinding wheelhead assembly. The surfaces of the columns shall accurately fit mating surfaces of the base and shall be bolted to the base forming single unit construction of the columns and base or the columns shall be cast integrally with the base. Means shall be provided to align the columns with the rotary worktable to meet the alignment tolerances given in table II. The columns shall contain and guide cross rail counterweights, when furnished. The columns shall accept the cross rail elevating mechanism housing. An access door(s) or removable plate(s) shall be provided to service components housed within the columns, as applicable.

3.3.5 Cross slide. The machine shall have a cross slide to support and guide the precision transverse movement of the grinding wheelhead assembly along the horizontal guideways of the cross rail (X axis). Unless otherwise specified (see 6.2.1), the cross slide shall have rapid traverse and automatic feed in either direction over the entire length of the cross rail. Interlocks shall be provided to prevent simultaneous operation of the power rapid traverse and the power feed systems. Means shall be provided for manual infeed of the grinding wheelhead assembly. Cross slide feed shall be controlled on type I machines by the operator, on type II machines by the automatic cycle, and on type III machines by the CNC system. Means shall be provided to lock the grinding wheelhead assembly after positioning. Grinding wheelhead cross feed travel and rate shall meet the requirements given in table I.

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3.3.6 Grinding wheelhead assembly slide. The grinding wheelhead assembly shall be precision fitted to the cross slide guideways, rigidly mounted on a compound slide allowing swivel motion (for grinding tapers) and vertical reciprocation of the grinding wheelhead assembly. The grinding wheelhead assembly shall traverse vertically in the slide guideways. All machines shall be provided with both rapid traverse and automatic reciprocation of the grinding spindle between adjustable stops. The vertical slide shall have down feed for the entire length of the stroke. An operator-adjustable dwell period shall be provided at both ends of the grinding stroke on type I machines. Automatic cross feed of the grinding wheelhead in adjustable increments shall be provided at both ends of the vertical stroke for feeds in either direction. The grinding wheelhead assembly vertical traverse and reciprocation shall be manually controlled by the operator on type I machines, controlled by the automatic cycle on type II machines, and controlled by the CNC system on type III machines. Means shall be provided to allow face grinding. When specified (see 6.2.1), impulse feed with adjustable feed increments shall be provided to feed into an internal or external diameter without reciprocation of the grinding wheelhead.

3.3.6.1 Grinding wheelhead assembly slide swivel motion. Swivel motion shall be accomplished through a worm and worm gear, adjustable by a handwheel or similar manual means. When specified (see 6.2.1), type II machine swivel motion shall be controlled by the automatic cycle. When specified (see 6.2.1), swivel motion shall be controlled by the CNC system on type III machines. Means shall be provided to lock the grinding wheelhead assembly into position after swiveling. A scale, graduated in 1 degree or less increments, shall be furnished to indicate angular grinding wheelhead assembly position.

3.3.7 Grinding wheelhead assembly. The grinding wheelhead assembly shall be fitted to the cross slide guideways, mounted on the compound slide, and shall include a wheelhead housing, spindle, grinding wheel, spindle drive unit, and a drive unit(s) for wheelhead assembly swivel, vertical positioning, and reciprocation, as applicable. Grinding wheelhead assembly vertical stroke, feed range, swivel, spindle speed, dwell, and motor horsepower shall meet the requirements given in table I. Unless otherwise specified (see 6.2.1), both grinding heads on style B machines shall be characteristically identical.

3.3.7.1. Grinding wheelhead assembly spindle. Unless otherwise specified (see 6.2.1), the grinding wheelhead shall house a cartridge type precision spindle which shall be rigidly fixed and attached to the wheelhead assembly slide. When specified (see 6.2.1), the spindle shall swivel $\pm$ 90 degrees. The grinding wheelhead spindle shall be driven by an electric motor through belts and pulleys. The spindle shall accept the driving mechanism at one end and a grinding wheel and holder at the other end. The grinding spindle shall be mounted in antifriction bearings designed to minimize axis and radial play and shall be dynamically balanced. Continuous or permanent lubrication to the spindle shall be provided.

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3.3.7.2 Manual grinding wheel downfeed. Manual downfeed of the grinding wheel into the work shall be accomplished using a handwheel or push button. Both coarse and fine feed shall be available with a graduated indicator or digital readout (see 3.1.1.2) provided to monitor wheel position.

3.3.7.3 Automatic grinding wheel downfeed. Each type II machine shall be equipped with automatic downfeed of the grinding wheel controlled by the automatic cycle (see 3.3.12). Means shall be provided for controlling feed and limiting the total downfeed for the overall grinding cycle. Downfeed rates and ranges shall be as specified in table I.

3.3.7.4 Computer numerically controlled wheelhead assembly downfeed. Each type III machine shall be provided with computer numerically controlled downfeed of the grinding wheel. Means shall be provided to control downfeed rate. Programmed feed stop points shall control downfeed cycle. Downfeed rates and ranges shall be as specified in table I.

3.3.8 Automatic wheel change. When specified (see 6.2.1), type III machines shall be capable of automatic wheel changing. Wheel change shall be actuated by command from the numerical control. Wheel change shall be accomplished by means of a power-operated mechanism which shall either interchange the wheel in the spindle with a wheel selected from the magazine or remove the wheel from the spindle and then insert the wheel from the magazine. Wheel change shall be possible any time during the operating cycle. The magazine shall permit convenient loading and unloading of wheels. The wheel changer shall be arranged for random wheel selection which shall include the capability of reusing a wheel that has been returned to the magazine after a programmed operation.

3.3.9 Wheel dresser. An adjustable wheel dresser, complete with commercial diamond, capable of dressing the face and periphery of the grinding wheel, shall be provided on each machine. The dressing cycle shall be manually operated on type I and type II machines and controlled by the computer numerical control system on type III machines. When specified (see 6.2.1), type III machines shall be equipped with a different wheel dresser for radius, angle, single point contouring, or rotary diamond form dressing, as required.

3.3.9.1 Dress-modes. On type III machines, programmable wheel truing cycles shall include, but not be limited to, dressing at the start of each cycle, a single interruption at the completion of rough grinding, or a combination of both; multiple dress; or dress on a skip cycle basis. In-resident programming shall establish appropriate dress rate based on desired dress lead and entered wheel speed. Dress modes shall be selective from sequence to sequence. Time for dress during automatic mode shall be as specified (see 6.2.1).

3.3.10 Grinding wheels. Unless otherwise specified (see 6.2.1), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All wheels shall be properly guarded.

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3.3.11 Automatic sizing. When specified (see 6.2.1), an automatic sizing system shall be provided on type I and type II machines. The system shall give continuous measurement of the work during grinding and visual indication the work is approaching size. When the work reaches a pre-set dimension, feed shall stop, the machine shall spark out, and the wheel shall retract. The cycle shall be initiated by the operator after inputting predetermined machine and workpiece parameters and loading the workpiece.

3.3.12 Automatic cycle. Type II machines shall be provided with an automatic cycle. Automatic cycling shall consist of complete part grinding without operator intervention from load to unload of part after predetermined machine parameters are established. The automatic cycle shall operate from a single control. The automatic cycle shall include, but not be limited to, the following functions:

a. Starting table rotation.
b. Starting grinding wheel rotation.
c. Engaging coolant pump.
d. Engaging power feed at pre-set rate until stopped by the automatic cycle.
e. Accomplishing pre-set timed sparkout.
f. Retracting wheelhead to pre-set location.
g. Stopping the grinding wheel rotation and coolant pump.
h. Stopping table rotation.

The automatic cycle shall be capable of being interrupted and re-entered during running.

3.3.13 Computer numerical control with manual data input. Type III machines shall be programmable through microprocessor-type computerized numerical controls (CNC) with full contouring capability and manual data input (MDI). The machine shall be two axes (Z and X) or three axes (Z, X, and C) programmable machines, as specified (see 6.2.1). The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, two or three axes, as applicable, positioning and feedrate control, wheelhead speed and direction of rotation, rapid traverse rates, axis dwell times, dress type, rate, and frequency, and shall initiate action should a machine fault occur. The control shall have linear and circular interpolation and be capable of two or three axes simultaneous control, as applicable. The axes under CNC control shall be identified in accordance with EIA Standard RS-267. Control features shall include a programmable interface between the MCU and grinder, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall accept part program data entered manually from a keyboard (MDI) and from an external input device compatible with the part program storage media (cassette tape

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cartridge, punched tape conforming to EIA Standard RS-227, or magnetic card) of the type specified (see 6.2.1). All necessary executive diagnostic and interface program routines shall be furnished as appropriate for the input media specified for controls having volatile memory. The control unit shall conform to EIA Standard RS-281. A line voltage of ±10 percent from normal shall not adversely affect the numerical control function. The control shall function in ambient temperatures ranging from 50 to 110 degrees F and relative humidities ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be 0.00005 inch for each linear axis and not more than 0.001 of a degree for each rotary axis. Positioning repeatability shall be 0.0001 inch.

3.3.13.1 Control features. Unless otherwise specified (see 6.2.1), the numerical control shall be provided with the following standard features:

a. An optical data display with a capacity of 256 characters (8 lines, 2 characters per line) capable of displaying block display, feedrate and override conditions, control and system error messages, and actual and desired position of each controllable axis indicated in a minimum of 7 digits with a positive or negative sign, as applicable.

b. Manual, automatic, single block, MDI, sequence number search, and optional stop control system modes of operation.

c. Absolute and incremental input capability.

d. Switchable inch/metric capability.

e. Part program storage in either solid-state or bubble-type memory equivalent to 250 feet of postprocessed part program tape data.

f. Standby battery power of not less than 72 hours.

g. Feed rate override.

h. Data input format and codes complying with EIA Standard RS-274 with automatic recognition of EIA Standards RS-244 (EIA) and RS-358 (ASCII) for punched tape part program storage media, when applicable.

i. Incremental and continuous jog for each axis.

j. Data edit keylock.

k. A control panel allowing, at least: Emergency stop, cycle start, operational stop, mode selection, feed hold, feedrate override, spindle speed override, and EOB stop.

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l. Program tryout feature to permit testing new part programs without machine motions.

m. EIA Standard RS-232 interface.

n. Other features, as required by the procuring activity.

3.3.13.2 Postprocessor. When specified (see 6.2.1), type III machines shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall convert specified input information to the form and format required to operate the grinder and control. The postprocessor shall be written for use on the type computer specified.

3.3.14 Drive mechanisms. Unless otherwise specified (see 6.2.1), the following drive mechanisms shall position each moving member in all axes of motion. Manual positioning shall be accomplished utilizing handwheels or pushbuttons and precision anti-backlash lead screws or ballscrews and nut assemblies for linear axes; movement shall be initiated and controlled by the operator. Power movement shall be accomplished by solid-state servo drives, electric motors, and precision leadscrews or precision, preloaded, antifriction recirculating ballscrew and nut assemblies. Computer numerically controlled machines shall also be equipped with position and velocity feedback devices forming a closed loop servo system. All leadscrews or ballscrews shall be hardened and precision ground to provide alignment and positioning accuracy, and be of a size, class, and precision suitable for the application. Leadscrews or ballscrews and nuts shall not exert excessive loads or cause premature deterioration of the feed drive motors or components. Feed rates and positions are to be monitored on computer numerically controlled machines by a feedback system and transmitted to the control.

3.3.15 Lubrication system. A lubrication system shall be provided to ensure adequate lubrication for all moving parts. When specified (see 6.2.1), the lubrication system shall be completely automatic. The system shall deliver lubricant to all mating gears, sliding parts, and bearings, except the sealed-for-life type. Each lubricant reservoir shall have a device for indicating and maintaining fluid level. The lubricant reservoir shall operate for a minimum of 72 hours of feed time without being serviced. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Interlocks shall be provided on the wheelhead and the table or carriage to prevent operation in case of pressure loss.

3.3.16 Coolant system. A coolant system shall be provided on each machine. The coolant system shall be either mist or flood type, as specified (see 6.2.1). A coolant delivery nozzle(s) and valve(s) shall permit the operator to direct and control the amount of coolant flow to the work and flush the base. The coolant reservoir shall be of sufficient capacity to permit full-flow coolant. The coolant system shall have means for draining and shall include a baffle or strainer, easily removable for cleaning. The reservoir shall have means for emptying and cleaning. Type I machine coolant

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systems shall be manually controlled; type II machine coolant systems shall be controlled by the automatic cycle (see 3.3.12); and the coolant system on type III shall be computer numerically controlled.

3.3.16.1 Flood coolant system. The flood type system shall include a sump or reservoir, a power driven pump, and piping. When specified (see 6.2.1), a coolant filtering system shall be provided with flood coolant.

3.3.16.2 Mist coolant system. The mist type system shall include a reservoir, filter, means for pressurizing the coolant, and means for directing and controlling the coolant.

3.3.17 Hydraulic system. If a hydraulic system is provided, the hydraulic system shall meet the requirements of JIC H-1 and be complete including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall ensure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gage to indicate the fluid level. The fluid temperature shall be maintained within the proper operating range during all normal operations of the machine. Means shall be provided to drain and clear out the hydraulic reservoir if the reservoir is not removable.

3.3.18 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless other specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt (with an allowable line fluctuation of $\pm$10 percent), 3-phase, 60-hertz (Hz) circuit. Initial voltage shall be as specified (see 6.2.1). Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating. Control enclosures shall meet the requirements of ANSI/NEMA ICS 6.

3.3.18.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG 1 and Section 430-7 of ANSI/NFPA 70.

3.3.18.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

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3.3.18.3 Solid state components. Solid state design shall be used throughout for electronic components.

3.4 Size and capacity. Unless otherwise specified (see 1.2 and 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine type, class, style, and size specified.

3.5 Performance. The machine shall be capable of performing as specified herein. When tested according to the procedures given in 4.7.1 and 4.7.2, the machine shall grind internal and external diameters round to within 0.0003 inch, straight to within 0.0002 inch per foot, and provide a surface finish not rougher than 8 microinch.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested using the procedures given in 4.7.3.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special wrenches normally furnished by the manufacturer.

b. One set of grinding wheels normally furnished by the manufacturer (not less than 3 wheels).

c. One worklight.

d. Table splash guard(s).

e. One extra set of grinding spindle belts and pulleys.

3.8 Optional equipment. Optional equipment and accessories shall be furnished as specified and described (see 6.2.1).

a. Additional worklight(s) for style B machines.

b. Grinding wheels: Number of each size, type, dimensional characteristics and composition shall be as specified (see 6.2.1), with required number of wheel holders, as applicable.

3.9 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

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3.10 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface letters on a contrasting background. All plates shall be corrosion-resistant.

3.10.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not larger than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, and machine alignment and accuracy test results) is required it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.4).

b. Acceptance test (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all quality conformance inspections, tests, and examinations (see 4.4) shall be performed in the manufacturer's designated indoor test area under normal environmental conditions and all acceptance test requirements (see 4.5) shall be performed only after the grinder is installed at its final location.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1 and 4.7.4, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.5 Acceptance test. An acceptance test shall be performed on the grinder to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be perform only after the grinder is installed at its final location and shall consist of the examination in 4.6 and the tests in 4.7.1 through 4.7.3. Failure of the grinder to pass one or more tests or any part of a test will be cause for rejection.

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4.6 Examination. The machine and related equipment shall be examined for compliance with all requirements of this specification, including compliance with all referenced standards.

4.7 Tests. Each machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the manufacturer. The measuring instruments used shall have evidence of calibration traceable to the National Bureau of Standards (NBS). At the option of the procuring activity, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1).

4.7.1 Operational test. The machine shall be operated at no load for not less than 4 hours. The machine and the control systems on types II and III machines shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. During the warm-up period, proper operation of all controls, motors, adjustment mechanisms, and accessories shall be verified. Should a malfunction of the automatic cycle on type II machines or the numerical control on type III machines occur during testing, it shall be corrected and the operational test repeated until a full 4 hours of running time is completed without failure.

4.7.1.1 Type II machines. After initial warm-up for not less than 4 hours, type II machines shall also be cycled continuously under the automatic cycle feature to verify its ability to receive and perform all cycle functions without error. This operation shall include table and wheel rotation, traverse of all slides, simultaneous movement of slides, feed cycles, and cycle stop, as applicable.

4.7.1.2 Type III machines. Each type III machine shall be cycled after the initial 4 hour warm-up period under numerical control to verify its ability to receive and perform all program functions without error. This operation shall include spindle speed and feed rate changes that include the highest, intermediate, and lowest settings of each range, rapid traverse of all slides, simultaneous movement of slides, and automatic feed cycles, as applicable. The numerical control system shall be further tested to verify proper operation of MDI, program edit, and system diagnostics.

4.7.2 Performance tests. Each machine shall be tested as follows to demonstrate its ability to finish grind internal and external surfaces and to ensure that its performance complies with the requirements of this specification.

4.7.2.1 Wheel dressing and truing test. On type III machines, wheel dressing and truing cycles are numerically controlled (see 3.3.8) and shall be tested before performing the grinding test in 4.7.2.2. The methods used in performing this test shall be the manufacturer's option provided they accurately determine the ability of the components and control unit to properly dress and true the grinding wheel.

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4.7.2.2 Grinding test. The machine shall be tested to demonstrate its capability to finish grind internal and external surfaces of a test part of AISI 1045 steel. The part shall have not less than a 6-inch diameter bore. Bore length shall be equal to the maximum depth of ground hole as given in table I for the machine size specified.

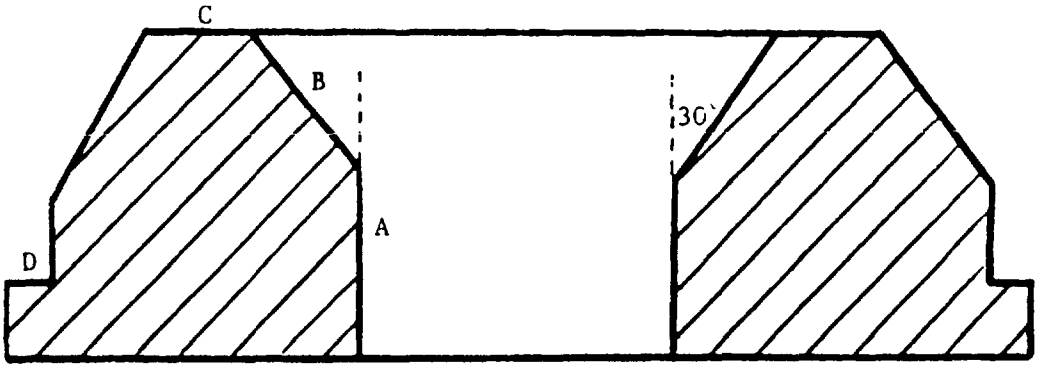
a. Grind a straight internal bore of the diameter determined by the manufacturer (not less than 6 inches).

b. Grind a 30 degree internal taper.

c. Face the end of the workpiece.

d. Grind an external shoulder of a size determined by the manufacturer.

All operations shall be performed in one set-up of the test workpiece. Finished physical dimensions shall meet the requirements of 3.5.



Surface A: Internal straight bore.
Surface B: Internal 30° taper.
Surface C: Face.
Surface D: External shoulder.

FIGURE 1. Sectional view of cylindrical performance test part.

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4.7.3 Alignment test. Each machine and its components shall be tested for alignment. Unless otherwise specified (see 6.2.1), the alignment requirements shall be in accordance with 3.6. The instruments and methods used in performing the tests shall be at the option of the supplier, provided they accurately determine the tolerances of the machine.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.2.11 using the test procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in shops for finish grinding of internal and external surfaces of workpieces with rapid metal removal rates to produce precision flat finishes on a wide variety of materials.

6.2 Ordering Data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, style, and size required (see 1.2).

c. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.1.1).

d. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.1.1.1).

e. Digital readout system, if not required; special features, if required (see 3.1.1.2).

f. Operator's controls, if different (see 3.1.4).

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g. Painting, if different (see 3.2.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.2.11).

i. Worktable T-slots, if different (see 3.3.2).

j. Cross rail elevating mechanism, if different (see 3.3.3.1).

k. Type III machine cross rail elevation CNC controlled, if required (see 3.3.3.1).

l. Cross slide traverse, if different (see 3.3.5).

m. Grinding wheelhead assembly impulse feed, if required (see 3.3.6).

n. Grinding wheelhead assembly swivel motion controlled by the automatic cycle on type II machines or the CNC system on type III machines, if required (see 3.3.6.1).

o. Style B machine grinding heads, if different (see 3.3.7).

p. Grinding wheel spindle, if different (see 3.3.7.1).

q. Swiveling spindle, if required (see 3.3.7.1).

r. Type III machine automatic wheel change, if required (see 3.3.8).

s. Type III machine wheel dresser, as specified (see 3.3.9).

t. Dress cycle time on type III machines, as specified (see 3.3.9.1).

u. Grinding wheels, if different (see 3.3.10).

v. Automatic sizing system, if required (see 3.3.11).

w. Axes under numerical control, as specified (see 3.3.13).

x. Part program storage media, as specified (see 3.3.13).

y. Numerical control features, if different (see 3.3.13.1).

z. Postprocessor, if required; specify computer name, model, form, and format (see 3.3.13.2).

aa. Drive mechanisms, if different (see 3.3.14).

bb. Automatic lubrication system, if required (see 3.3.15).

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cc. Mist or flood type coolant, as specified (see 3.3.16).

dd. Filtering system on flood coolant system, if required (see 3.3.16.1)

ee. Electrical system, if different; initial input voltage, as specified (see 3.3.18).

ff. Motor enclosure, if different (see 3.3.18.1).

gg. Size and capacity, if different (see 3.4).

hh. Additional performance requirements, if required (see 3.5.1).

ii. Standard equipment, if different (see 3.7).

jj. Specify and fully describe optional equipment, if required (see 3.8).

kk. Grinding wheel number, type, dimensional characteristics, composition, and required number of wheel holders, if required (see 3.8.b).

ll. Specify applicable repair parts, if different (see 3.9).

mm. Lubrication plate, if different (see 3.10.1).

nn. Nameplate, if different (see 3.10.2).

oo. Hour meter, if required (see 3.11).

pp. Inspection conditions, if different (see 4.3).

qq. Quality conformance inspection, if different (see 4.4).

rr. Acceptance test, if different (see 4.5).

ss. Test requirements, if different (see 4.7).

tt. Alignment requirements, if different (see 4.7.3).

uu. If preservation, packing and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

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6.3 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-G-80239A	MIL-G-80239B
New	Type I
Type I	Type II, Class 1
Type II	Type II, Class 2
New	Type III
Style A	Style A
Style B	Style B
Size 1612	Deleted
Size 2518	Size 1
Size 3618	Size 2
Size 4224	Deleted
New	Size 3
Size 5424	Deleted
New	Size 4
Size 6624	Deleted
Size 9024	Deleted

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing:

Abrasive wheel, grinding
Automatic cycle
Computer numerically controlled
Grinder, surface
Grinding machine, surface
Machine tool, grinding
Manual control
Rotary table, single
Surface grinder
Vertical spindle, double, single
Wheel, abrasive, grinding

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TABLE I. Sizes and capacities.

Characteristic	Types I and II				Type III			
	Classes 1 and 2				Classes 1 and 2			
	Styles A and B				Styles A and B			
	Size 1	Size 2	Size 3	Size 4	Size 1	Size 2	Size 3	Size 4
Capacity:								
Standard worktable diameter, inches	24	36	48	60	24	36	48	60
Maximum swing, inches	36	44	60	68	36	44	60	68
Maximum hole depth, inches	18	18	24	24	18	18	24	24
Vertical grinding head stroke, inches	21	21	27-1/2	27-1/2	21	21	27-1/2	27-1/2
Clearance over rotary table, inches	20	20	48	48	20	20	48	48
Clearance over work spindle, inches	23	23	54	54	23	23	54	54
Grinding head swivel, degrees, inboard	0 to 30				0 to 30			
outboard	0 to 45				0 to 45			
Feeds and speeds:								
Work spindle speeds, number	Infinite				Infinite			
Work spindle speed range, RPM	6 to 120		5 to 90		6 to 120		5 to 100	5 to 80
Grinding head rapid travers, IPM, vertical	2 to 80				0 to 100			
horizontal	0 to 40		0 to 30		0 to 100			
Grinding head cross feed range, IPM, either and both ends of stroke	0.0002 to 0.003				0.0001 to 0.050			
Grinding wheel reciprocation rate, IPM	5 to 100				5 to 100			
Grinding wheel spindle speed, RPM	3600		1800		3600		1800	
Grinding head tarry period, either and both ends of stroke, seconds	0 to 14				0 to 14			
Motor horsepowers:								
Grinding wheel spindle	10	10	15	15	10	10	15	15
Work spindle	3	3	5	5	3	3	5	5
Coolant	1/4	1/4	1/2	1/2	1/4	1/4	1/2	3/4
Hydraulic	5				3	3	5	5
Cross rail elevating mechanism*	1-1/2				1-1/2			

NOTE 1: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

NOTE 2: * When applicable.

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TABLE II. Alignment test requirements.

Test to applied	Allowable deviation
Rotary table top surface flat in all directions - up to 60" diameter - 61" and larger diameter	 0.0003"/12" 0.0005"/12"
Axial runout of rotary table top and periphery surfaces	0.0002" TIR
Crossrail parallel with rotary table top	0.0005"/12"
Crossrail vertical travel (class 2 machines), straight	0.001"/12"
Grinding head vertical travel, straight	0.0003"/12"
Grinding head spindle travel square with surface of rotary table	0.0005"/12"
Grinding head spindle square with surface of rotary table	0.0003"/12"
Taper of wheel spindle true running	0.0003" TIR
Wheel and work spindles, axial runout	0.0003" TIR

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - None
Air Force - 84
DLA-GS

User activities:
Army - AR
Navy - SH, AS
Air Force - None

Preparing activity:
DLA-IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80239B	Grinding Machines, Non-Traversing Rotary Table, Vertical Spindle, Universal, Manual, Automatic, and CNC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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